02 Identification of different celltypes in Seurat

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Introduction

This RMarkdown describes an analysis of the single-nucleus RNA-sequencing count matrix from GSE138852 using Seurat. The dataset consists of a count matrix containing raw UMI counts derived from snRNAseq of human entorhinal cortex nuclei. Quality control of the nuclei was already performed according to the information provided in the article by Grubman *et al.*, with low-quality cells removed during prior filtering steps.

The scope of this RMarkdown includes normalization of the count matrix, identification of highly variable features, and dimensionality reduction. Cell-type annotation is performed using BRETIGEA but is not covered within this RMarkdown.

Research question

For this project the main research question is *"How can a reliable and reproducible analysis pipeline be set up to identify cell type-specific differential gene expression in Alzheimer's disease using a filtered expression matrix generated from snRNA-seq data?"*. Initially the research question was aimed at creating the entire pipeline with raw sequencing data as a starting point however, the publicly available data was lacking information on cellbarcodes and UMIs. Therefore the expression matrix published for GSE138852 was used as a starting point.

The sub-question(s) that is answered in this RMarkdown is as followed;

- Can the most variable features be identified with the filtered expression count matrix as input?
- How should the count matrix be used, should the groups be separated (healthy vs. Alzheimers disease)?

Packages and Requirements

In order to run the analysis in Seurat a few packages need to be installed first. Packages and package versions are managed using `renv`, instructions for setting up the R project with `renv` are given in the README of the GitHub repository *ad_transcriptomics_dge*.

About the count matrix

The count matrix was downloaded from the GEO-database. The count matrix is a table which stores the gene expression measurements of all measured nuclei. Usually the matrix contains the raw expression measurements but in the case of GSE138852 some filters were already applied, the matrix contains;

- 13214 nuclei (before filtering 14876 nuclei were measured)
- Only nuclei with less then 10% of the total features being mitochondrial
- None of the features that are related to postmortem interval
- Only cells that are within the 5th - 95th percentile regarding gene and UMI count.

Loading the dataset and creating a Seurat object

The count matrix was stored in a comma separated values file (.csv-file). To be able to use this file for creating a Seurat object `read.csv()` was used to create a data.frame first. Then the data.frame was used as input for the function `CreateSeuratObject()`.

Pre-processing of the dataset

Since the count matrix had already been pre-processed as described in "About the count matrix", no additional pre-processing is performed at this point. However to check if the dataset was indeed pre-processed as stated in the article by Grubman *et al*. a column was added containing the percentage of mitochondrial features. A violin plot was generated and visualized in figure 1.

Normalization of the dataset

After normalizing the dataset `FindVariableFeatures()` was used to detect the most variable features. The selection method that was used is `vst`, and `nFeatures` was set at 2000. A scatter plot of the output of `FindVariableFeatures()` generated and visualized in figure 2.

Conclusion

Misschien dat dit result moet worden

In order to check if the count matrix contained the data as described by the article by Grubman *et al*. and the GEO-database (Accession number: GSE138852) the data was plotted in three violin plots (figure 1). These violin plots confirm that the filtering was applied as stated in the article by Grubman et al., most nuclei show between 400 to 800 detected features, the amount of total RNA molecules detected is between 500 - 1000 for most nuclei and the percentage of mitochondrial features is around 1,5% with no values >10%.

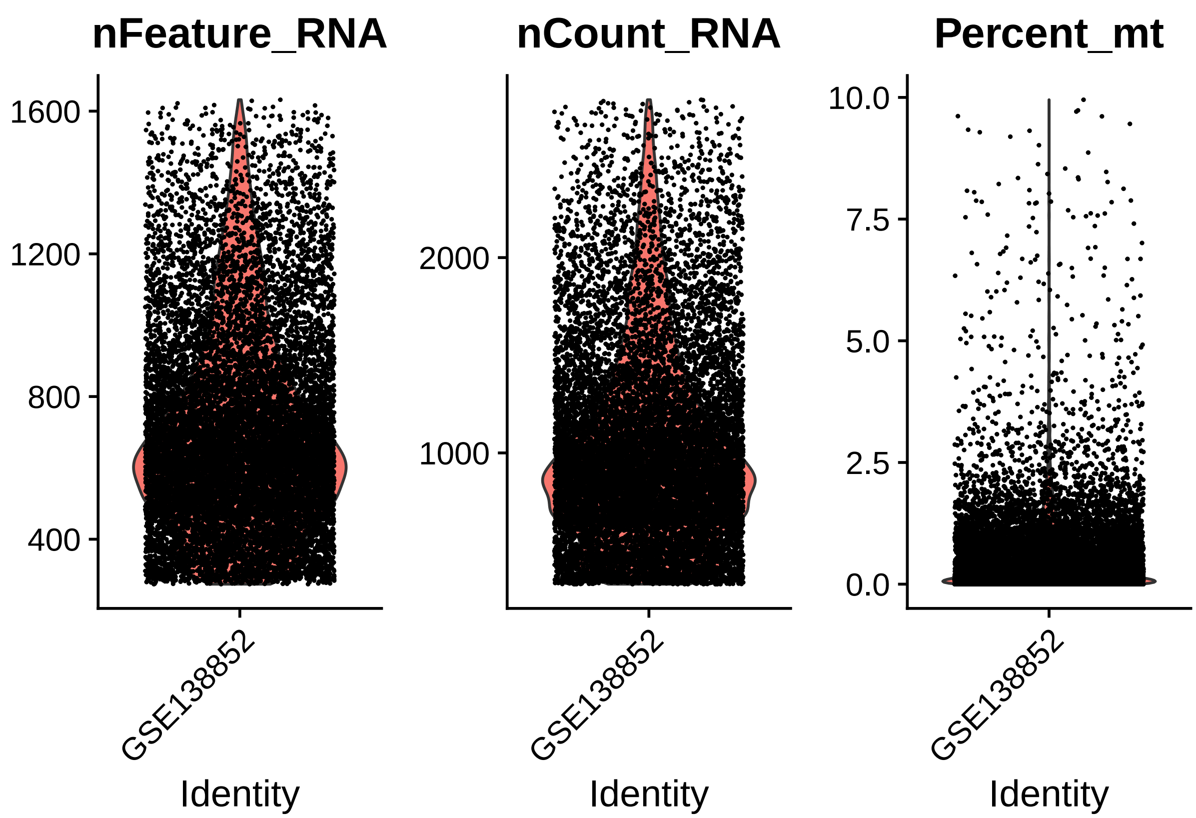

Figure 1: Violin plots of quality-control metrics. Shown are the number of detected genes per nucleus (nFeature_RNA), total UMI counts per nucleus (nCount_RNA), and the percentage of mitochondrial gene expression (Percent_mt). The distributions of these plots are as expected with prior quality-control filtering as described by Grubman et al.

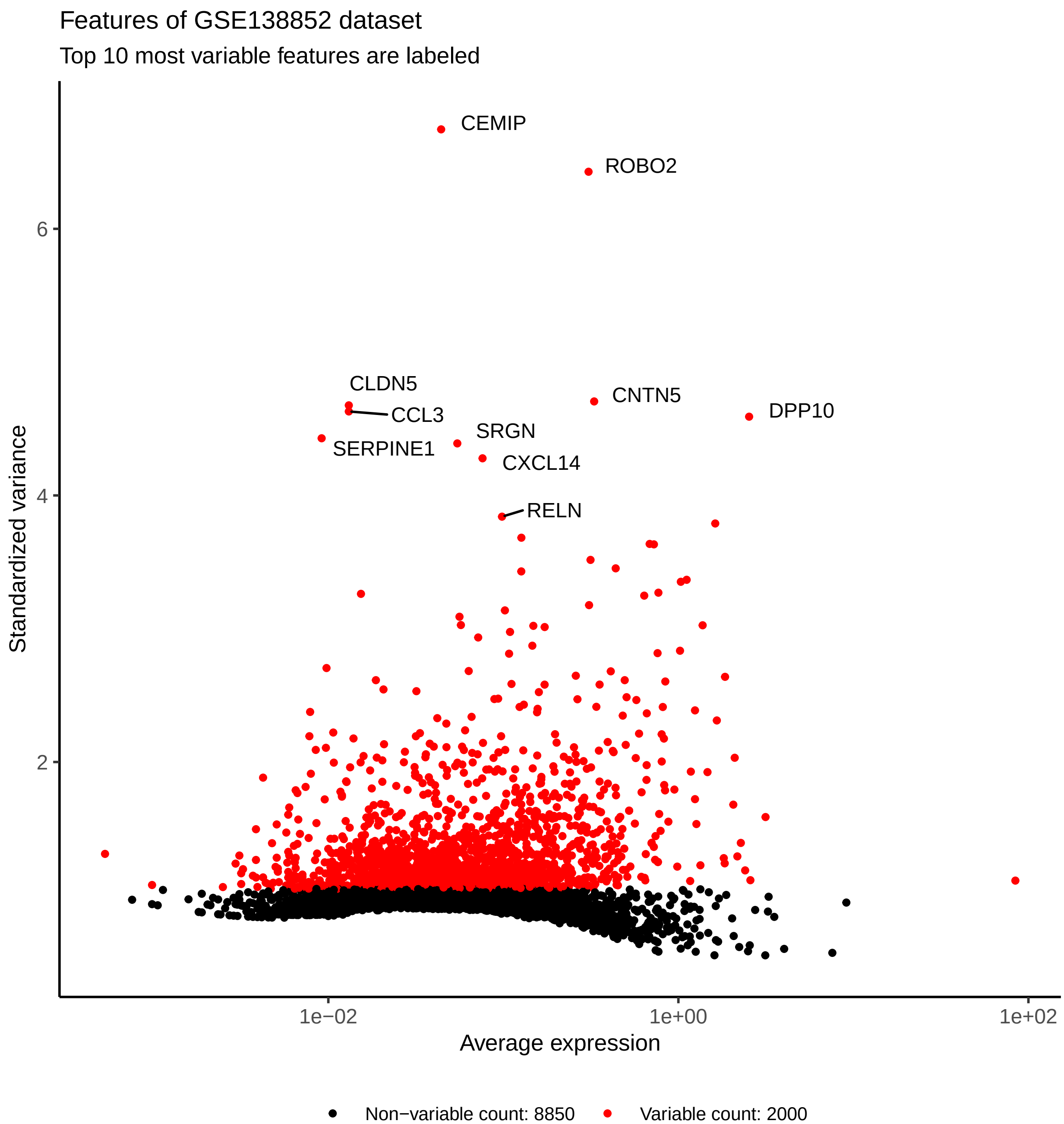

Figure 2: Scatterplot of variable features.